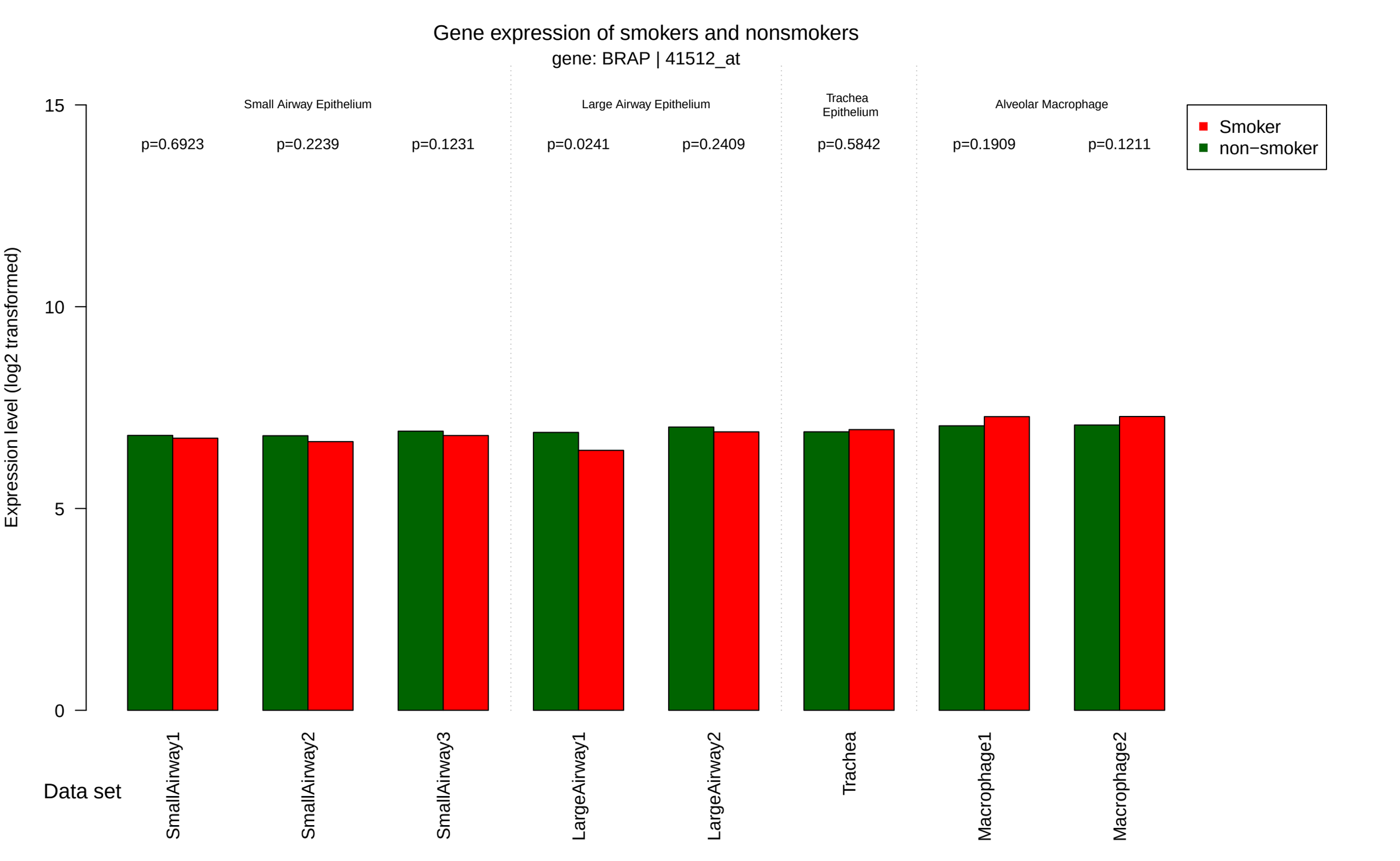
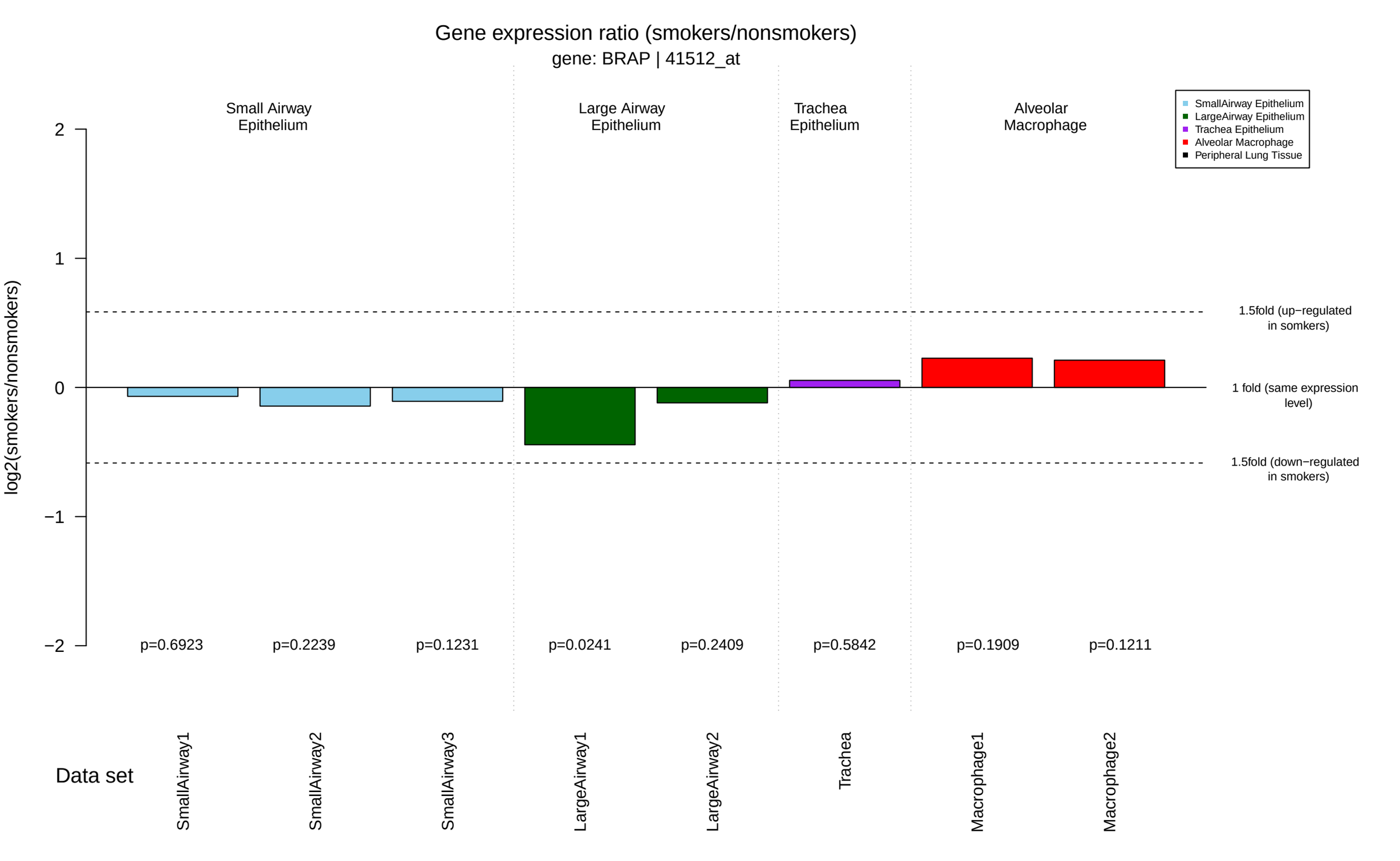
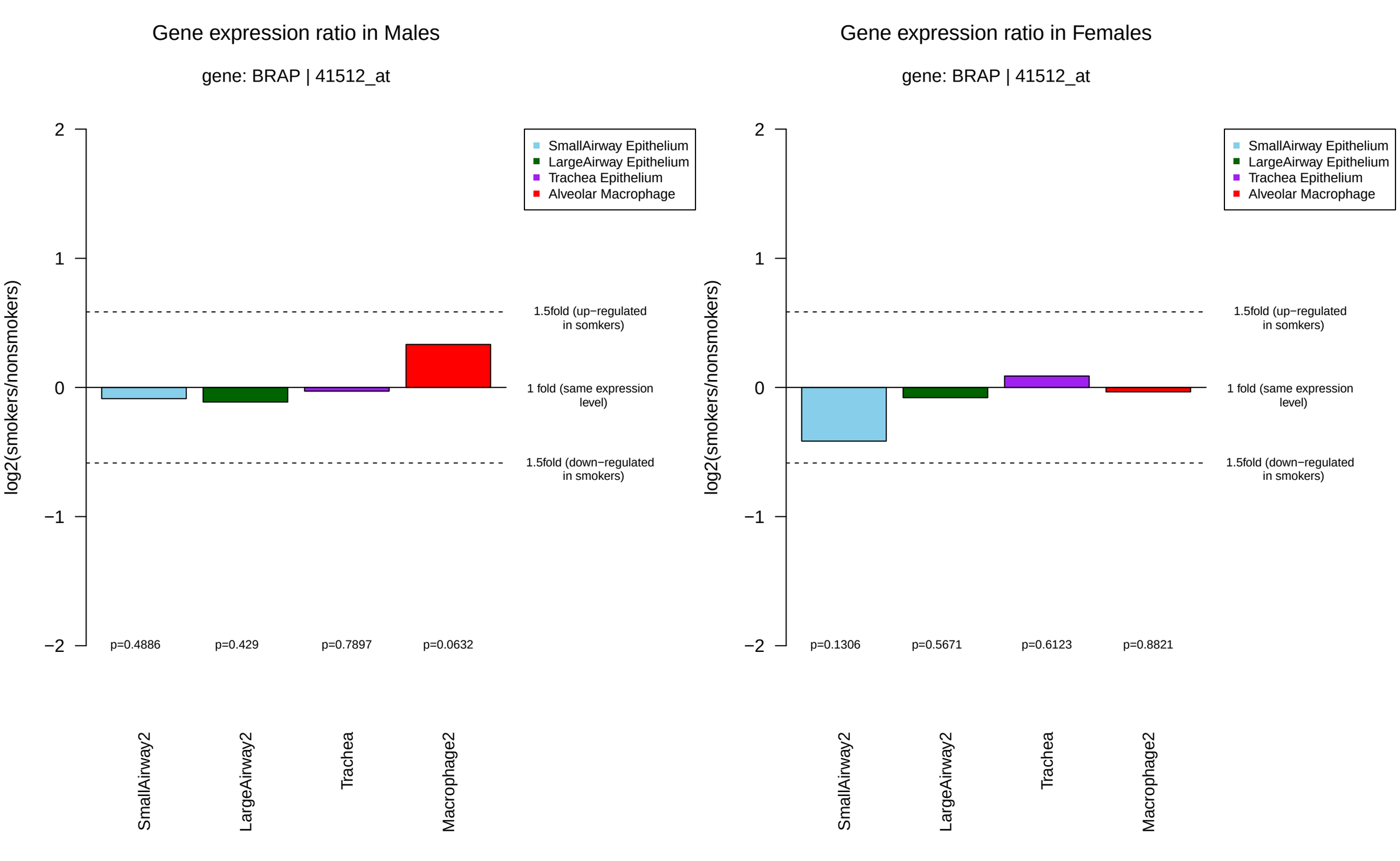
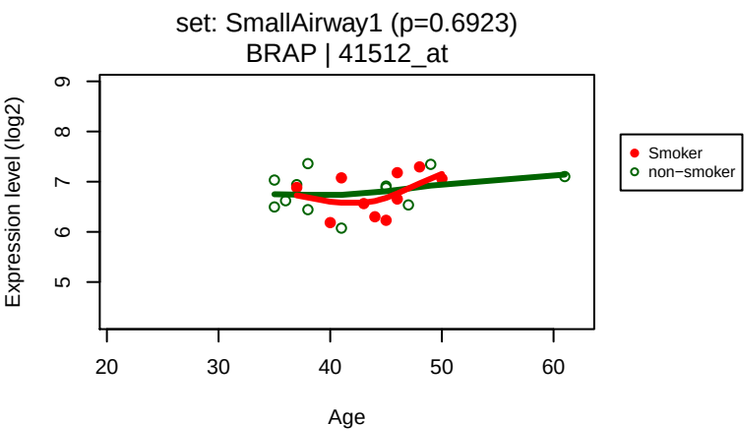
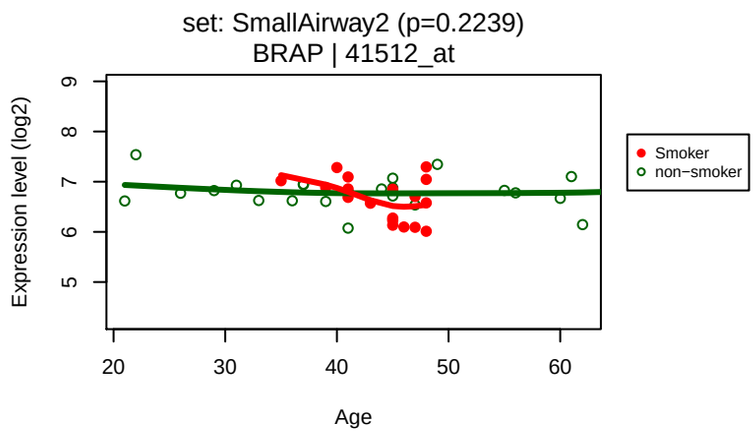
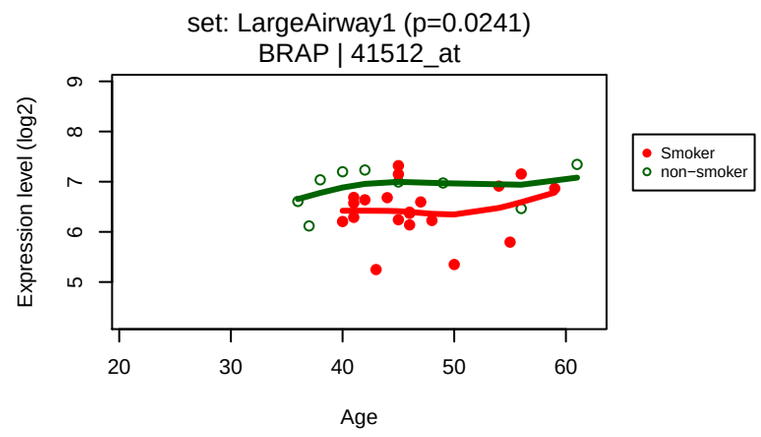

Gene expression of smokers and nonsmokers
gene: BRAP | 41512_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6923
p=0.2239
p=0.1231
p=0.0241
p=0.2409
p=0.5842
p=0.1909
p=0.1211
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

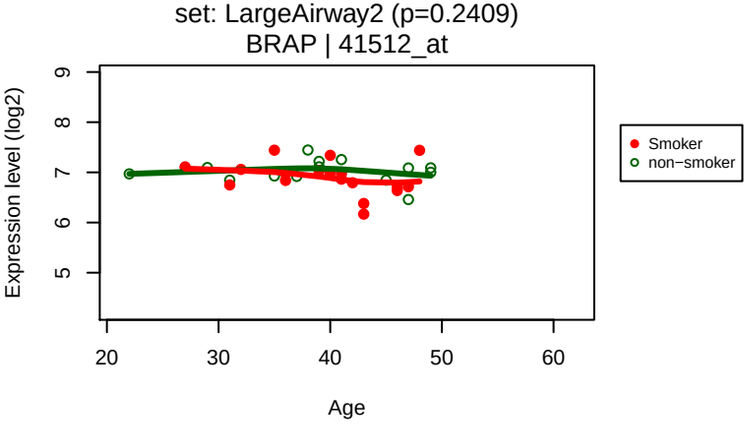

Gene expression ratio (smokers/nonsmokers)
gene: BRAP | 41512_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6923
p=0.2239
p=0.1231
p=0.0241
p=0.2409
p=0.5842
p=0.1909
p=0.1211
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

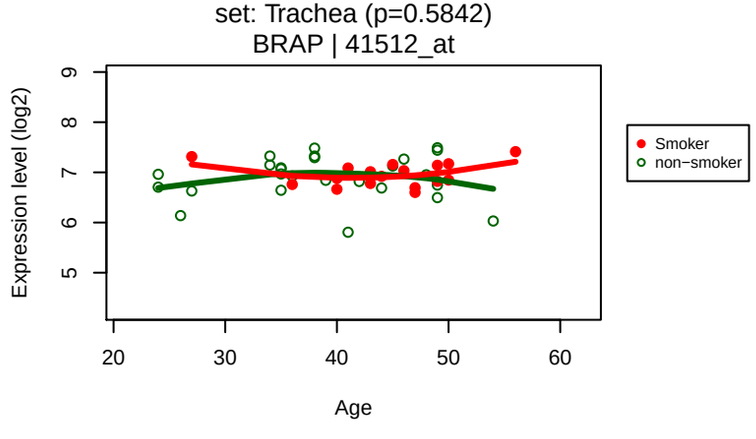

Gene expression ratio in Males
gene: BRAP | 41512_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4886
p=0.429
p=0.7897
p=0.0632
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRAP | 41512_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1306
p=0.5671
p=0.6123
p=0.8821
SmallAirway2
LargeAirway2
Trachea
Macrophage2

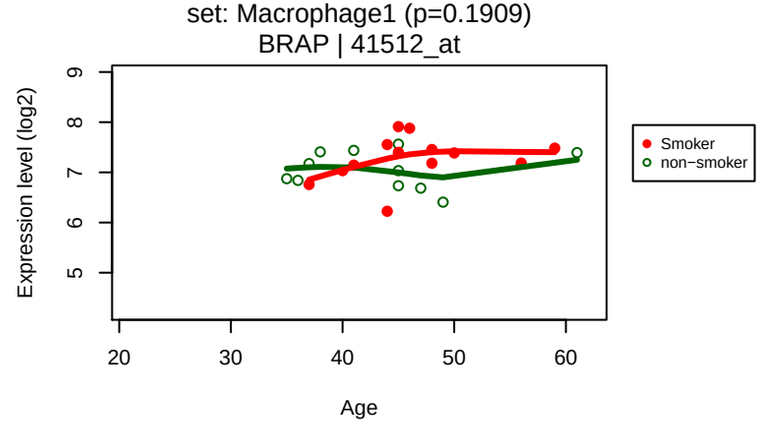

set: SmallAirway1 (p=0.6923)
BRAP | 41512_at
Expression level (log2)
Age
Smoker
non−smoker

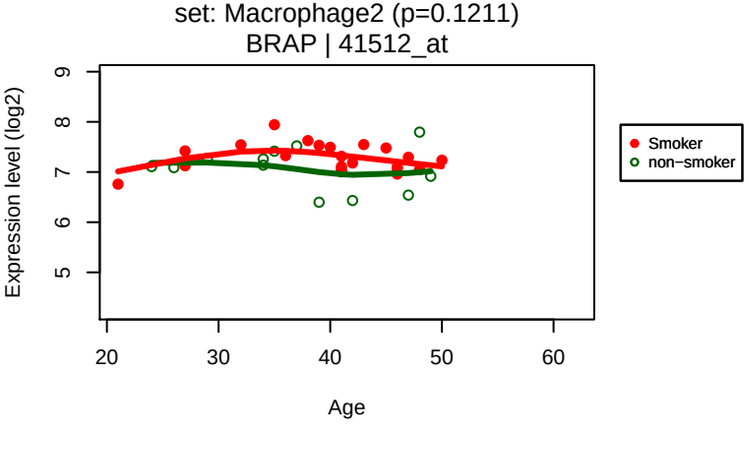

set: SmallAirway2 (p=0.2239)
BRAP | 41512_at
Expression level (log2)
Age
Smoker
non−smoker

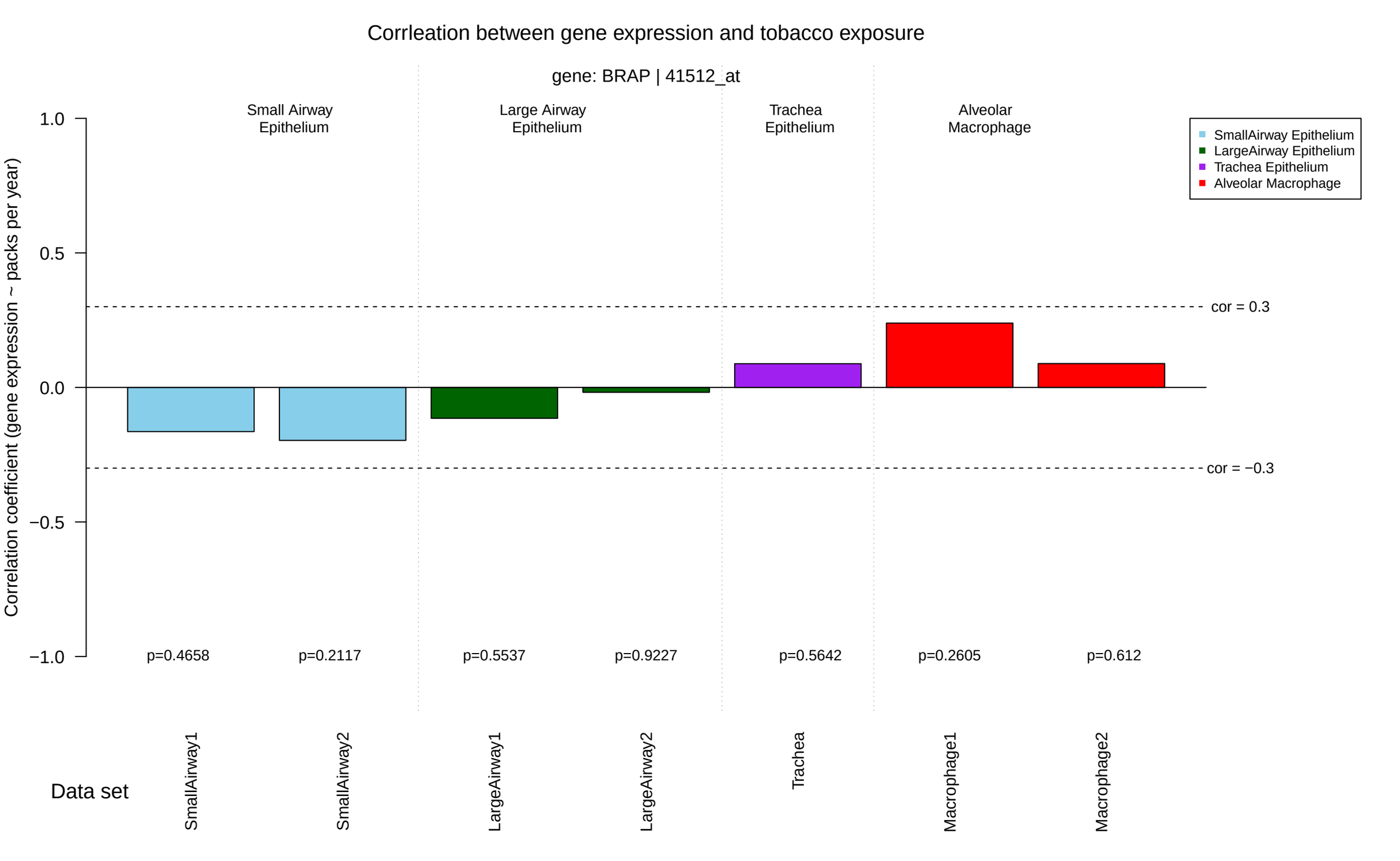

set: LargeAirway1 (p=0.0241)
BRAP | 41512_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2409)
BRAP | 41512_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5842)
BRAP | 41512_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1909)
BRAP | 41512_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1211)
BRAP | 41512_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRAP | 41512_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4658
p=0.2117
p=0.5537
p=0.9227
p=0.5642
p=0.2605
p=0.612
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2